

Accessible tools for deep learning in bioimage analysis

Showcase of DL tools a Python-capable microscopy user can use today

Zuzana Čočková - BioImage analyst at IMCF



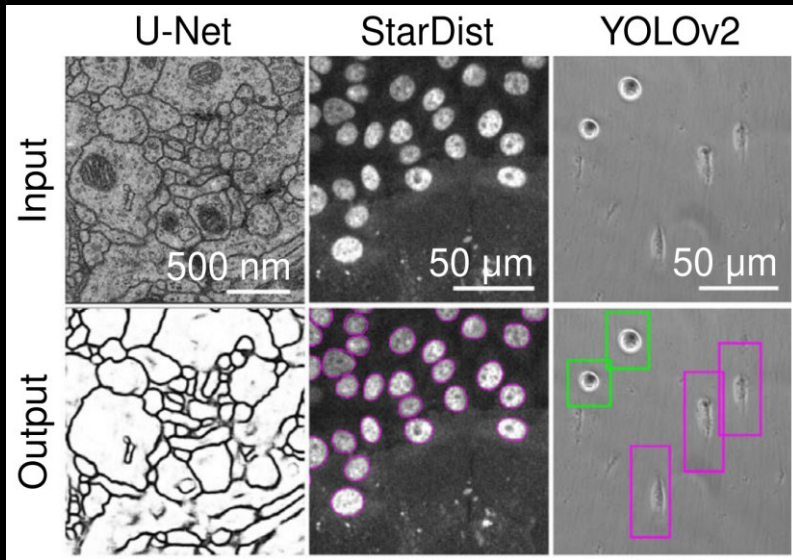
Imaging Methods
Core Facility



Faculty of Science
Charles University

DL technologies are powerful ...

Segmentation & Object detection



Restoration & Super-resolution

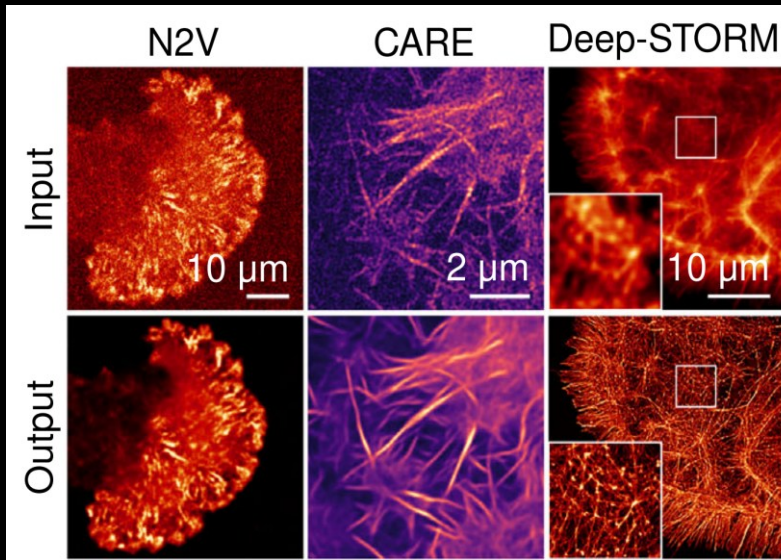
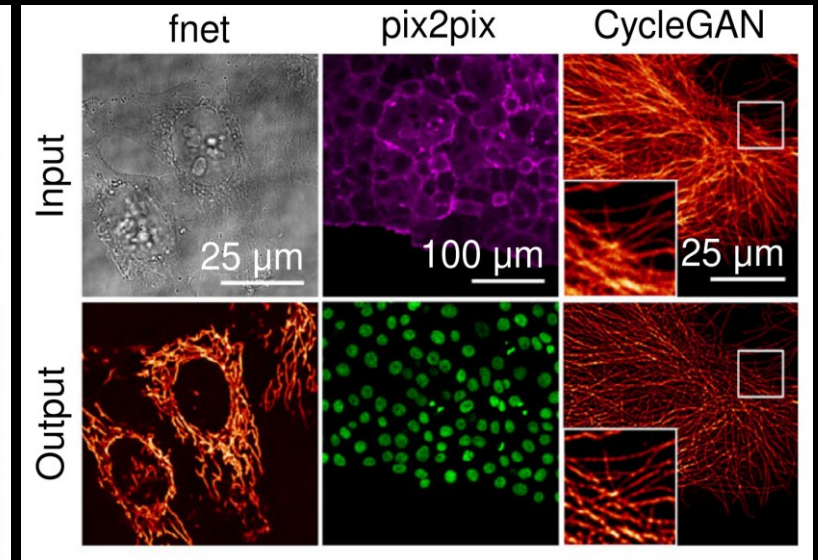
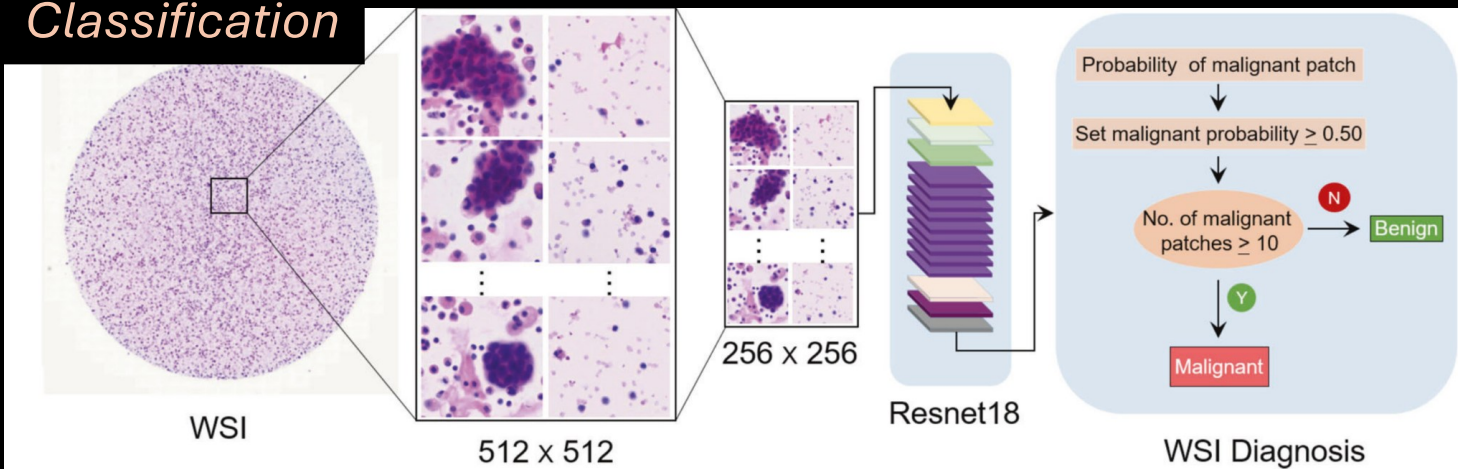


Image-to-image translation



Classification



ZeroCostDL4Mic: an open platform to use
Deep-Learning in Microscopy

<https://doi.org/10.1101/2020.03.20.000133>

Deep convolutional neural network-based
classification of cancer cells on cytological pleural
effusion images.

<https://doi.org/10.1038/s41379-021-00987-4>

... but can be hard to access for non-experts

👥 Community efforts are lowering the barrier



TOOLS EXIST

free to use
low-code setup
integrations

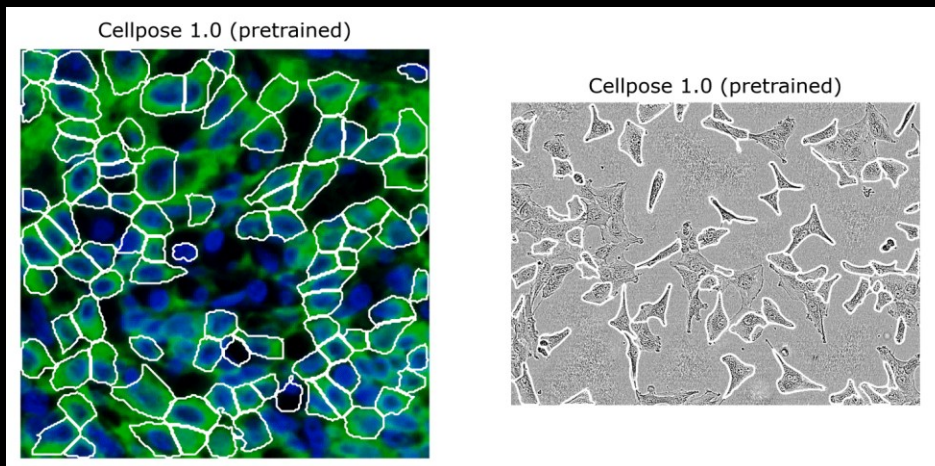
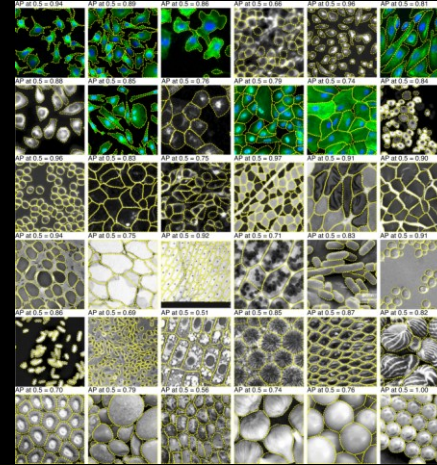
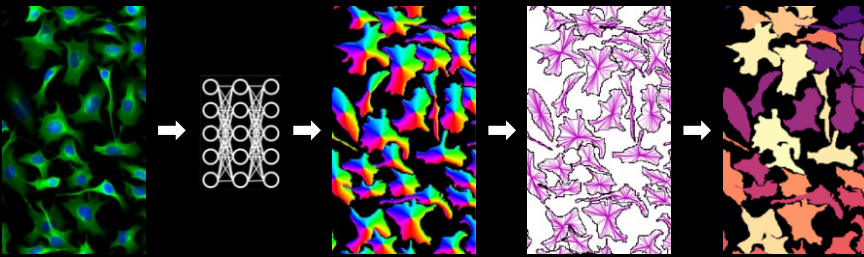
MODELS EXIST

pretrained
openly shared
interoperable



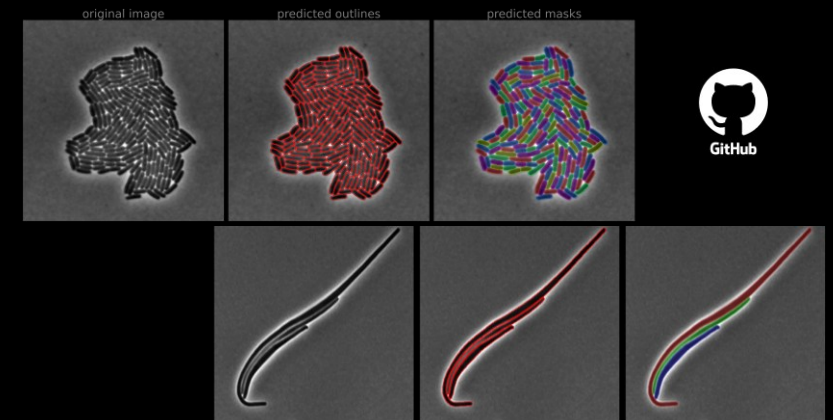
Cellpose (cell segmentation)

- Generalist cell segmentation algorithm
- Works on many modalities
- Available pre-trained models
- Re-training (fine-tuning) is possible



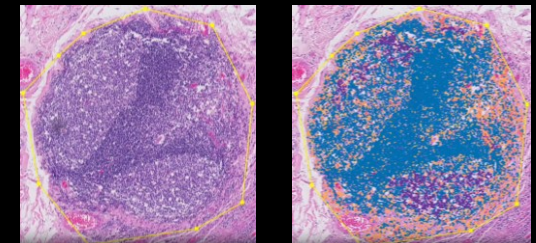
Omnipose

Better handling of elongated cells, irregular morphology, dense bacteria



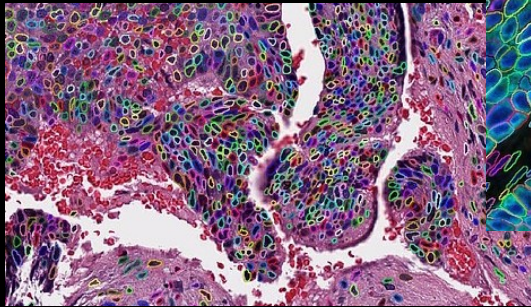
Classpose

Segmentation + semantic classification in whole slide images

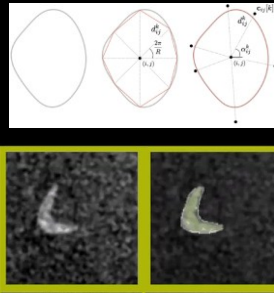


More ready-to-use segmentation solutions

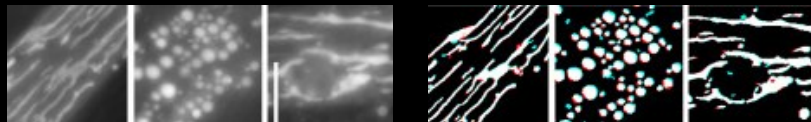
InstanSeg



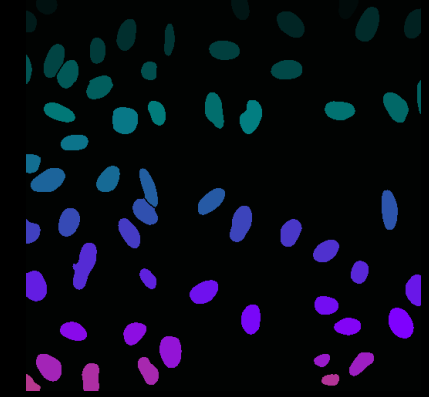
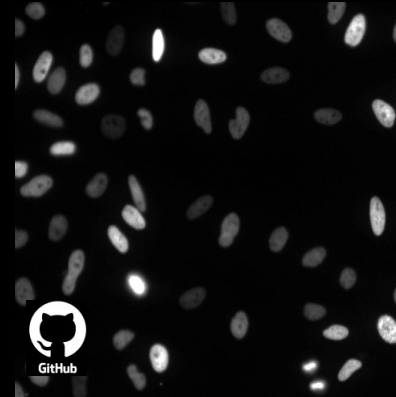
SplineDist



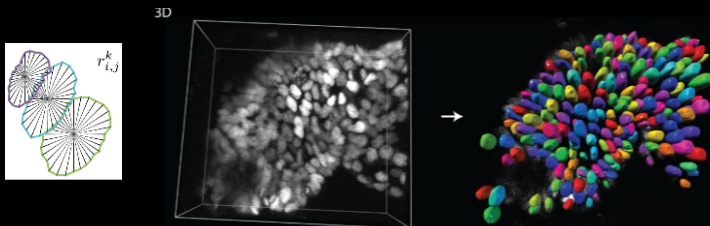
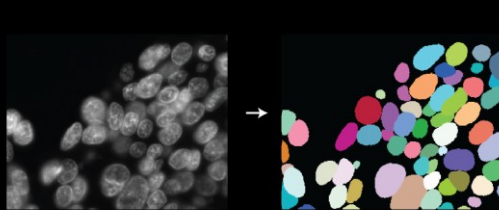
MitoSegNet



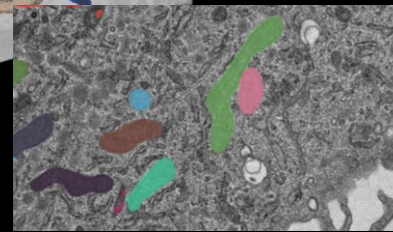
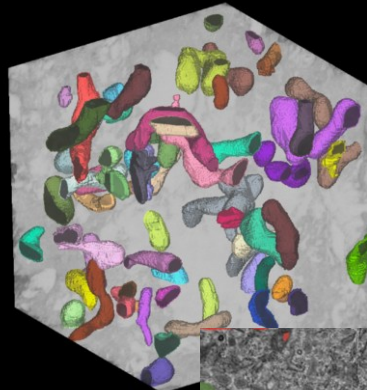
DeepCell



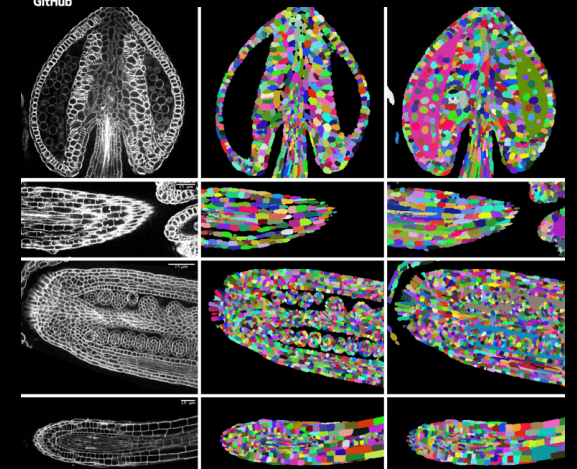
StarDist



MitoNet



PlantSeg



Segment Anything Model (SAM)

- model developed by Meta AI (ai.meta.com/sam3/)
- to identify and segment any object in an image
- is promptable
- Multiple adaptations for microscopy:
 - micro_sam (Segment Anything For Microscopy)
 - SAMcell
 - cellposeSAM

And many napari plugins...

napari-segment-anything | napari-segment-everything |
napari-sam | napari-sam3-assistant | napari-SAM4IS

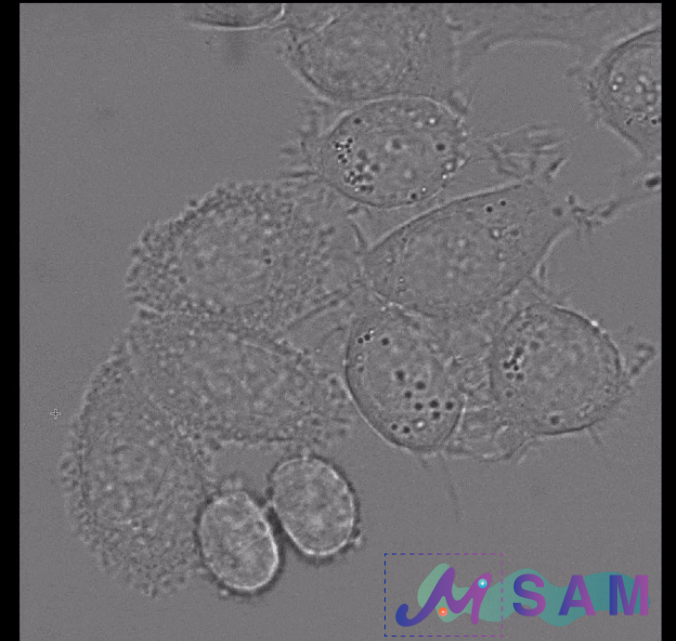
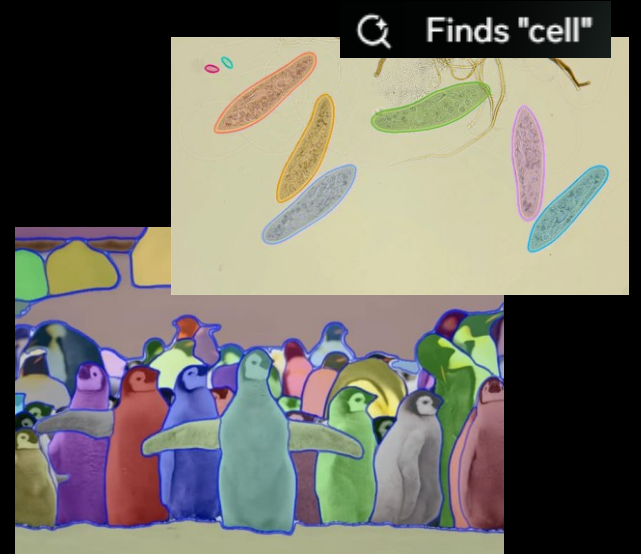
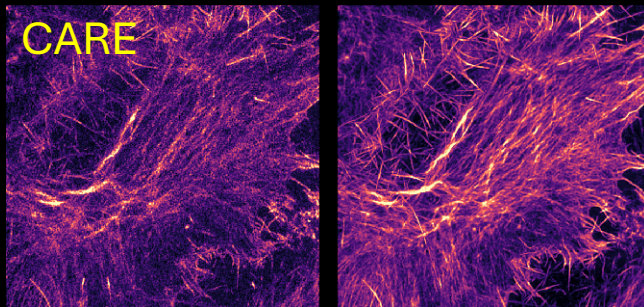


Image restoration

CAREamics  - a PyTorch library simplifying the use of supervised and self-supervised DL image restoration algorithms

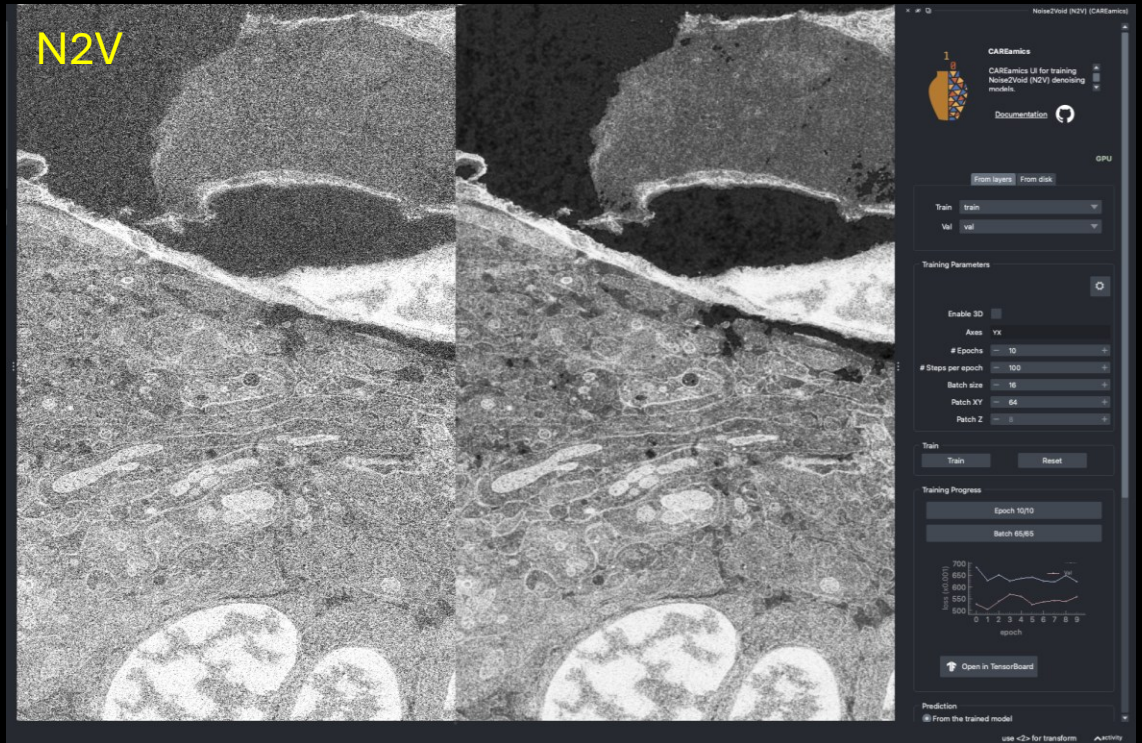
One library, many restoration algorithms

- Content-aware Image Restoration (CARE)
- Noise2Noise (N2N)
- Noise2Void (N2V, N2V2, structN2V)



low-res

high-res



DL platforms

ZeroCostDL4Mic | BiaPy

- Colab notebooks 
- broader DL landscape
segmentation, detection, image-to-image translation, classification, restoration, registration, ...
- train the model yourself
or provide compatible pretrained model

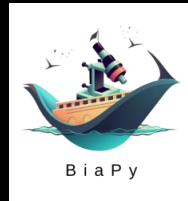
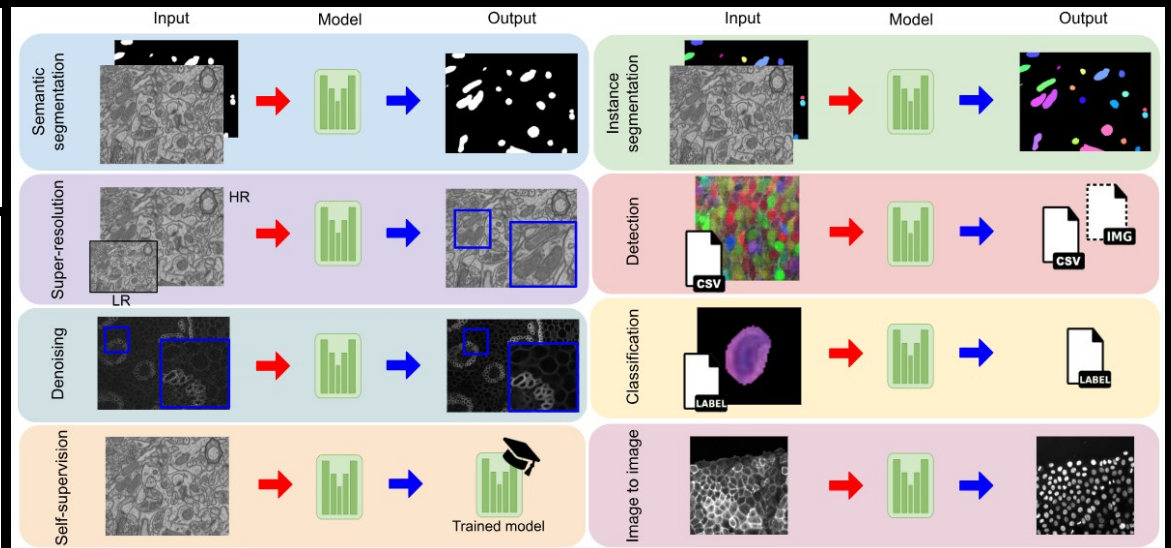
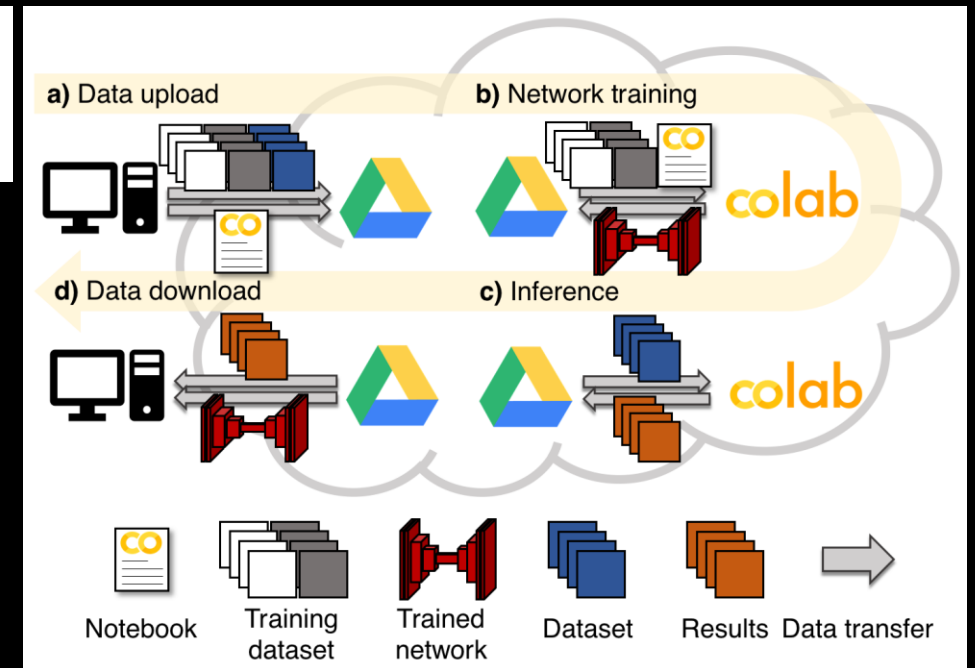


Image classification (2D)	Image denoising (2D)	Image to image (2D)	Instance segmentation (2D)	Object detection (2D)	Self-supervision (2D)	Semantic segmentation (2D)	Super-resolution (2D)
Image classification (3D)	Image denoising (3D)	Image to image (3D)	Instance segmentation (3D)	Object detection (3D)	Self supervision (3D)	Semantic segmentation (3D)	Super-resolution (3D)

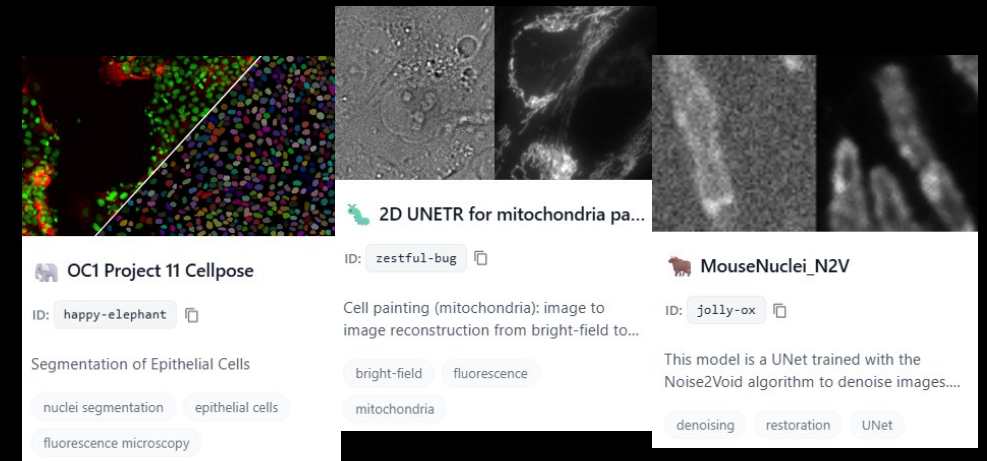


Model hubs (repositories)



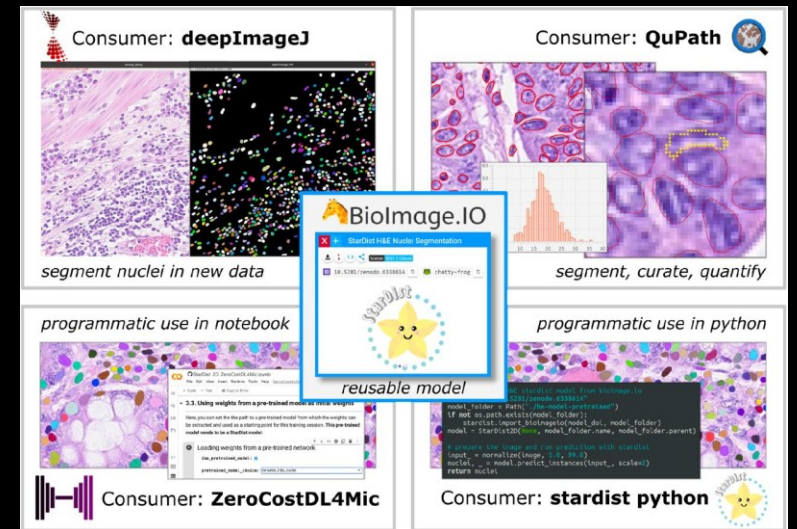
BioImage Model Zoo

- community-driven
- standardized models
- interoperable formats - ensures models work across partner tools
- covers a wide range of tasks

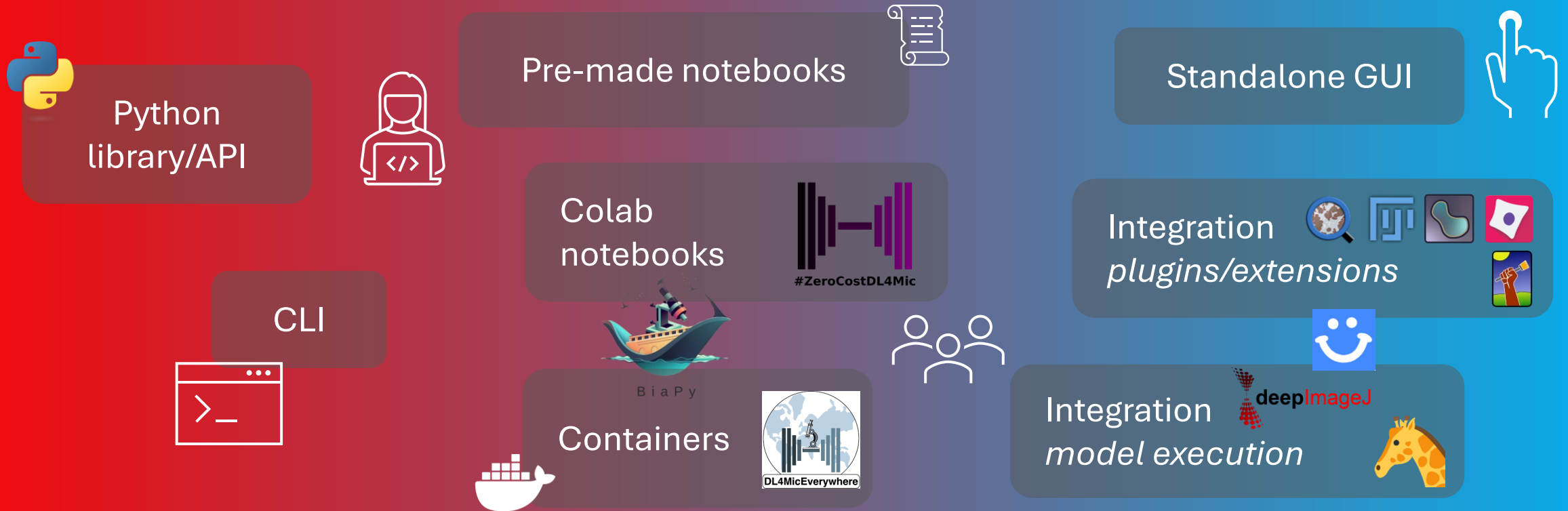


Hugging Face

- broader ML ecosystem
- growing number of biomedical and microscopy models



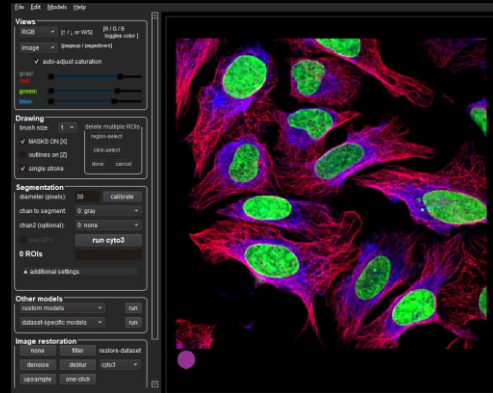
Interoperability - ways to use DL techniques



— programming/technical skills requirements —————→

Example: Cellpose ecosystem

1. via GUI



2. via Python code

```
from cellpose import models, io

image = io.imread('path/to/image')

model = models.Cellpose(model_type='cyto')

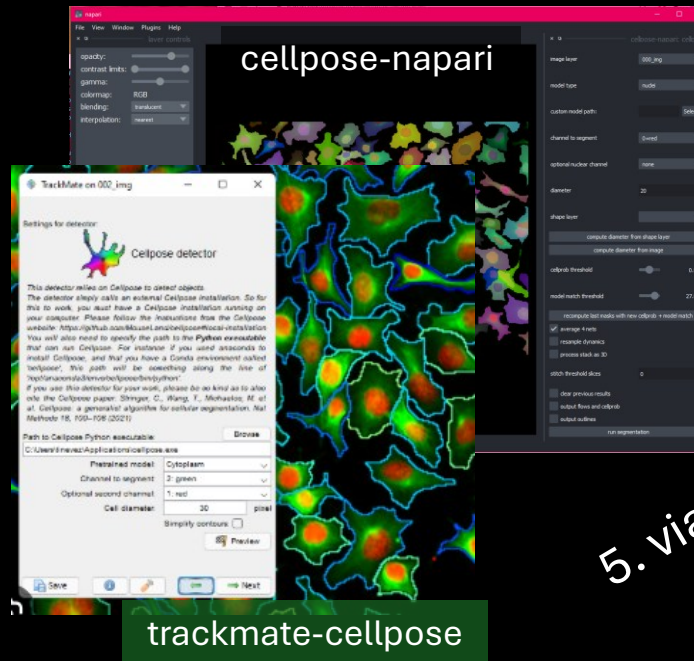
masks, flows, styles, diams = model.eval(
    image,
    diameter=30,
    channels=[0, 0]
)
```

3. via CLI

```
C:\Users\zuzka> cellpose --dir ./images --pretrained_model cyto3 --diameter 30 --chan 0 --save_png
```

4. via pre-made notebooks, Colab notebooks or containers

Cellpose (2D and 3D) here	Instance segmentation (Cells or Nuclei)	Fully supported	05/09/23 ✓ working (IH)	here ZeroCostDL4Mic	Open in Colab
---	---	-----------------	----------------------------	--	---------------



May have limited functionality

- limited training
- limited batch processing
- only 2D support
- reduced parameter control

5. via integrations

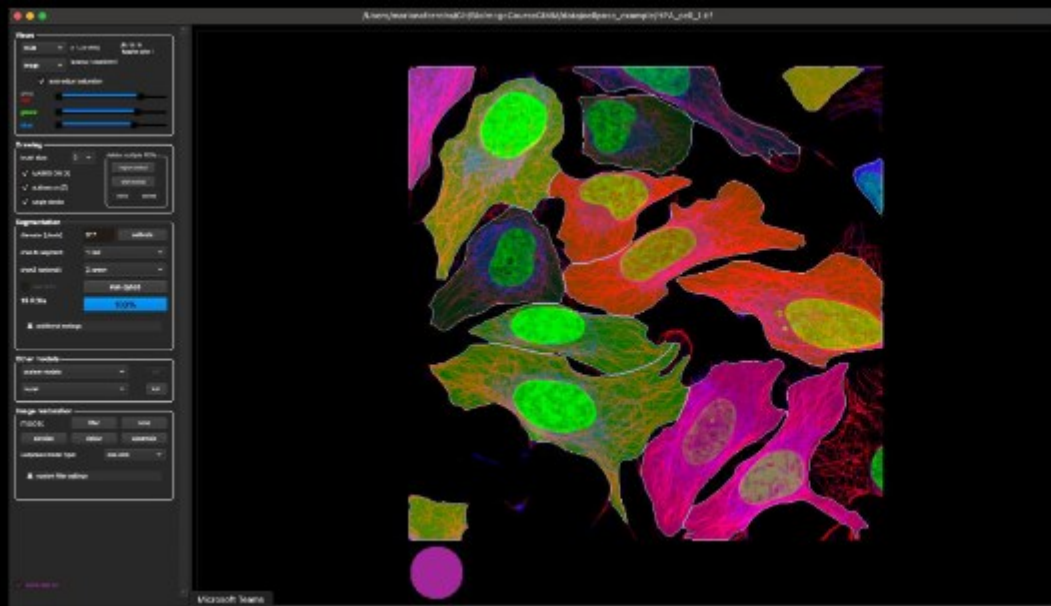


DL use case - Cellpose



Hands-on
program 

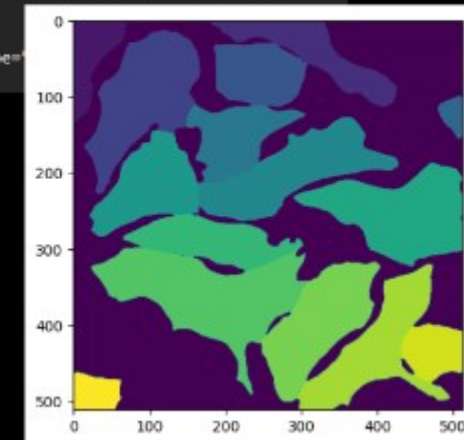
- Notebook #4
 - approach A: built-in GUI
 - approach B: via Python code (local notebook and Google Colab)



```
from cellpose import io, models
import matplotlib.pyplot as plt

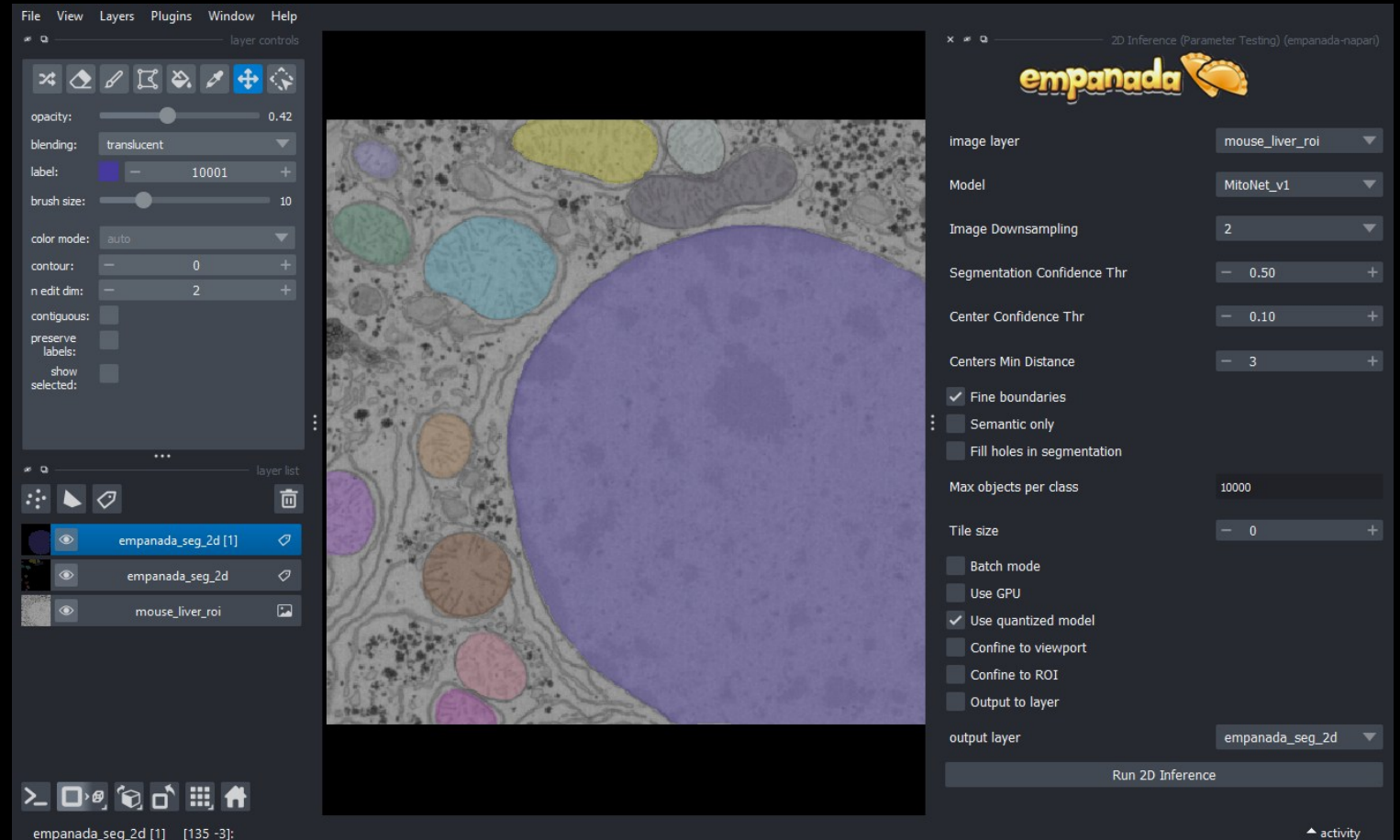
# import the image
img = io.imread('../data/cellpose_example/HPA_cell_1.tif')

# Initialize the model
model = models.Cellpose(model_type='
```



DL use case - empanada

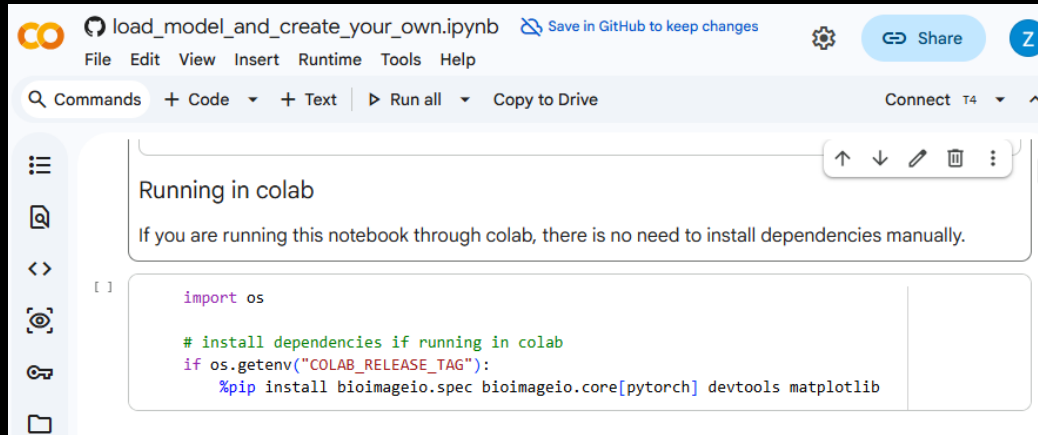
- Notebook #5
 - approach: napari



DL use case - BioImage Model Zoo

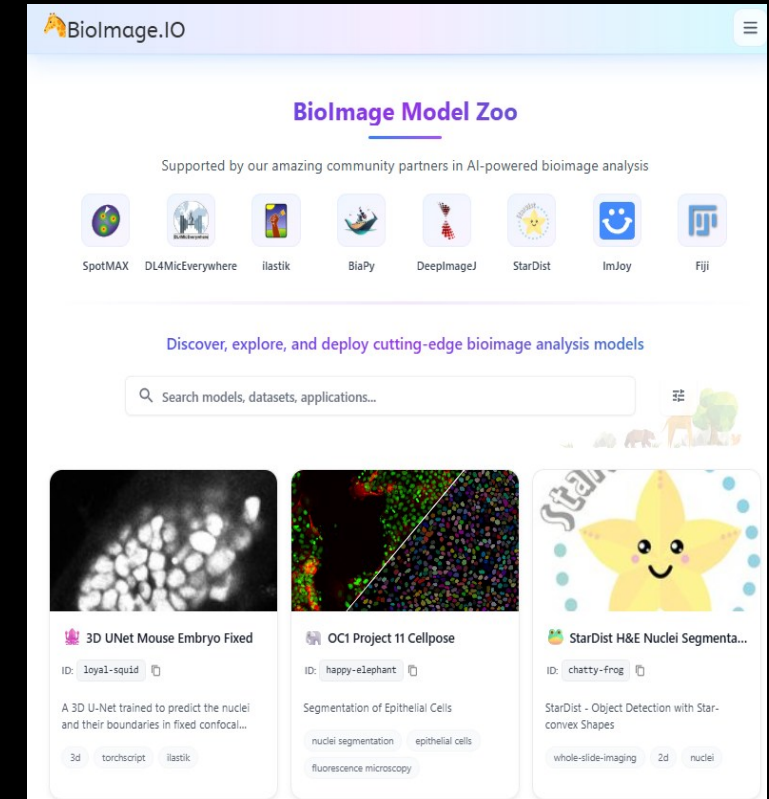
Hands-on
program 

- Notebook #6
 - demo via BioEngine
 - use via Python code in Google Colab



```
import os

# install dependencies if running in colab
if os.getenv("COLAB_RELEASE_TAG"):
    %pip install bioimageio.spec bioimageio.core[pytorch] devtools matplotlib
```



ambitious-sloth 

chatty-frog 

happy-elephant 

loyal-squid 

trustworthy-llama 

wild-pig 

serious-turtle 

